

LESSON 6-7 • SESSION 2 • PRACTICE • FORM A

Multiples and Least Common Multiple

Learning target – I can list multiples of two numbers and find their least common multiple. • 6.NS.B.4

REMEMBER

- A **multiple** is what you get when you skip-count by a number: 6, 12, 18, 24, ...
- The **least common multiple (LCM)** is the smallest multiple above 0 that two numbers share.
- Watch out: $8 \times 10 = 80$ is a common multiple, but it is **not always the least**.

WORKED EXAMPLE

A library van visits every 8 days and an art cart every 10 days. Both visit today. When do both visit again?

Step 1 Multiples of 8: 8, 16, 24, 32, 40.

Step 2 Multiples of 10: 10, 20, 30, 40.

Step 3 The first number on both lists is 40.

Answer LCM = 40, so in 40 days

WORDS **multiple** (*múltiplo*) • **common multiple** (*múltiplo común*) • **least common multiple** (*mínimo común múltiplo*)

PRACTICE

show your work • label units

1. Find the LCM of each pair.

a. 7 and 4 _____

b. 8 and 12 _____

c. 6 and 9 _____

d. 4 and 10 _____

2. **Two Buses.** A red bus reaches a stop every 15 minutes. A blue bus reaches the same stop every 20 minutes. Both are at the stop now.

a. Multiples of 15: _____ Multiples of 20: _____

b. Minutes until both buses are there again: _____ min

3. **Cups and Lids.** Cups come in packs of 14. Lids come in packs of 21. A store wants the same number of cups and lids.

Hint Skip-count by 21 and stop at a multiple of 14.

a. Fewest cups and fewest lids: _____

b. Packs of cups ____ Packs of lids ____

Show your work.

4. **Practice Days.** Band meets every 5 days. Choir meets every 3 days. Both met today. In how many days will both meet again?

Hint List the meeting days for each group.

Show your work.

Answer _____ days

5. **Owen's Shortcut.** Owen says the LCM of 16 and 20 is 320 because $16 \times 20 = 320$. Is he right? Give the LCM.

Hint Skip-count by 20 and test each number.

Show your work.

LCM _____

- 6. Two Machines.** Machine A stamps every 25 seconds. Machine B stamps every 30 seconds. They both stamped just now.

Hint Divide the LCM by each machine's time.

- a.** Seconds until they stamp together again: _____ s
b. Stamps by A ____ Stamps by B ____ (up to that time)

Show your work.

- 7. Three Lists.** Find the least common multiple of 2, 5, and 6.

Hint Skip-count by 6 and look for a multiple of 5.

Show your work.

LCM _____

- 8. Two Alarms.** One alarm rings every 9 minutes. Another rings every 12 minutes. Both ring at 7:00 a.m.

- a.** Find the LCM of 9 and 12. _____
b. When do both ring together next? _____
c. How many times do they ring together after 7:00 and up to 9:00 a.m.? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Gia's Work.** Gia listed multiples of 3 and of 5. She said the LCM is 30, the second number that appeared on both lists. Explain her mistake. Then give the correct LCM.

TEST PRACTICE

choose the best answer

- 10.** The LCM of 6 and 8 is –

A 2 **B** 14 **C** 24 **D** 48

- 11.** Which number is a multiple of 7?

A 1 **B** 17 **C** 27 **D** 35

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

108 Z	7:36 E	150 I	320 Q	36 L	15 L	42 U	80 T	60 M	30 P
#2b	#3a	#4	#5	#6a	#7	#8a	#8b		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____